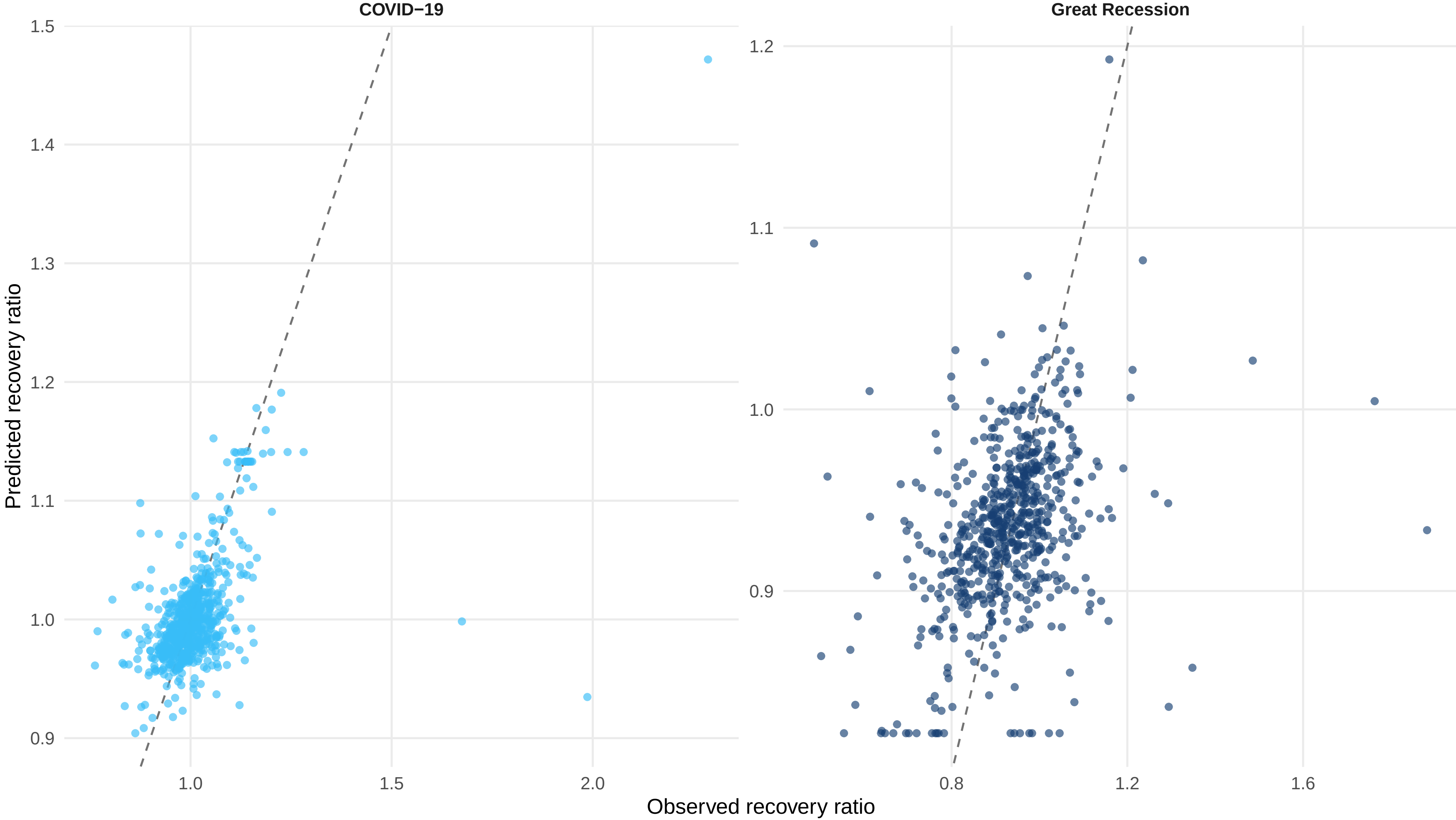
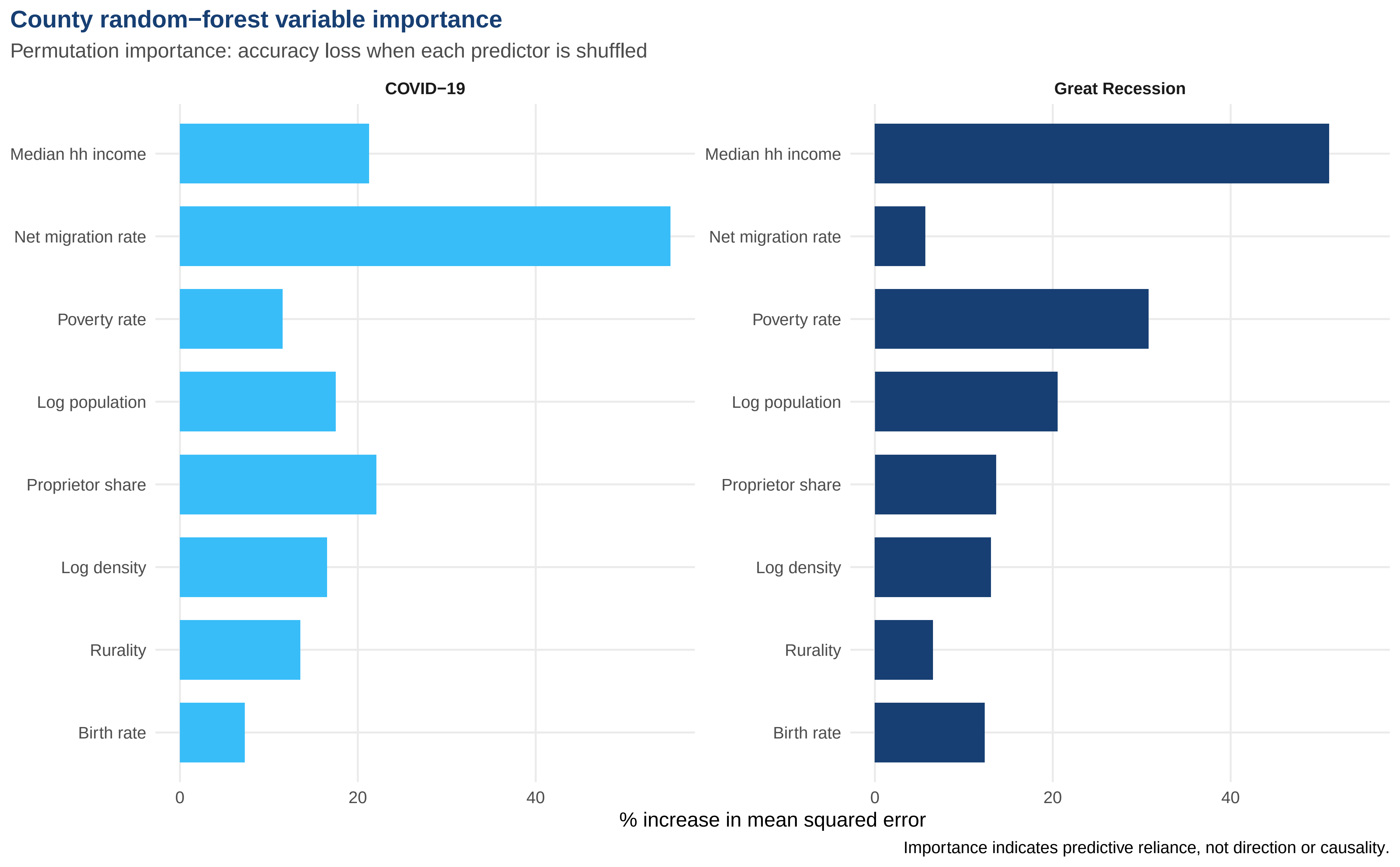
County random forest: observed versus predicted

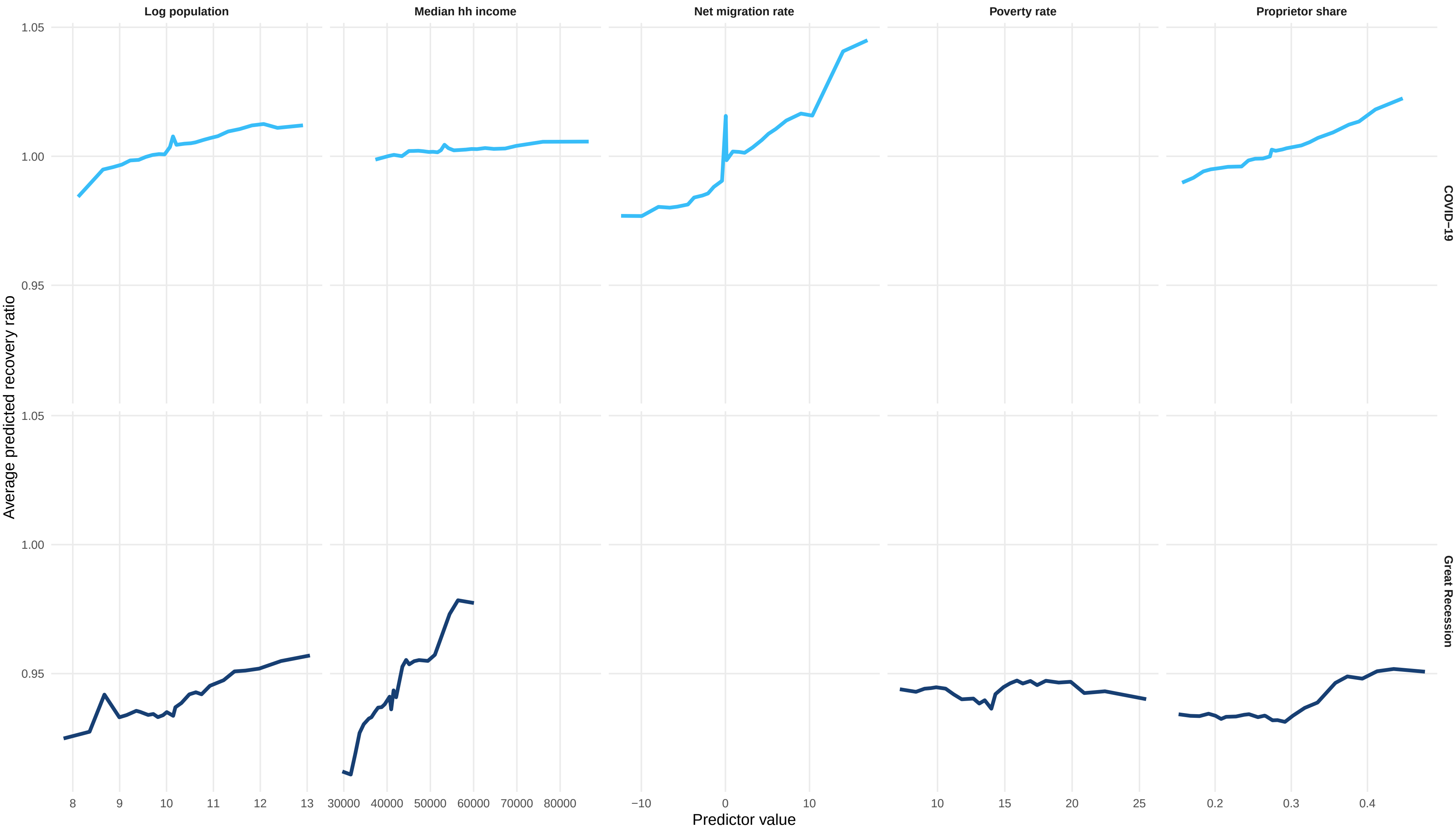
Predictions are from a reproducible 20% holdout set



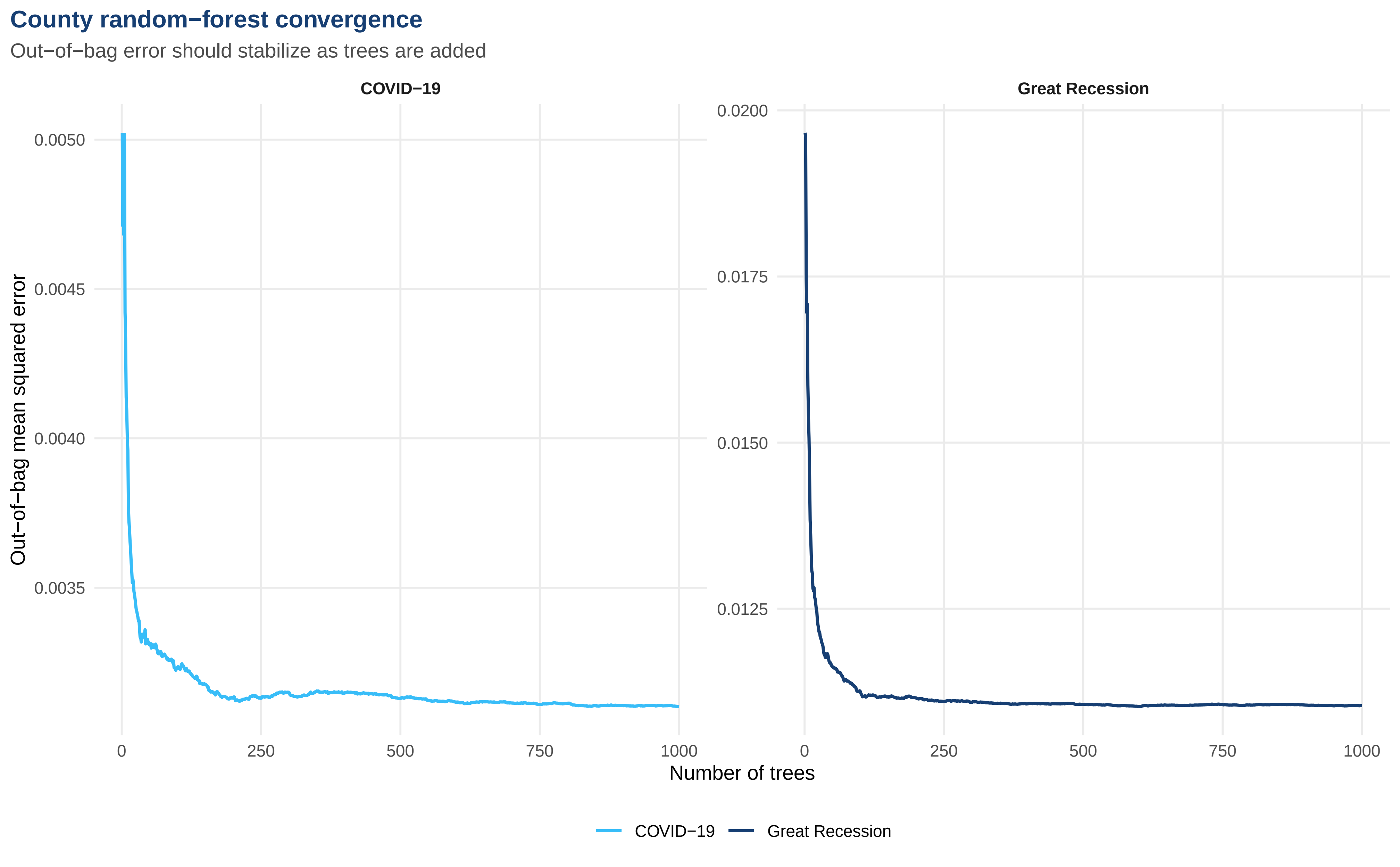

County random−forest variable importance
Permutation importance: accuracy loss when each predictor is shuffled
COVID−19
Great Recession
Median hh income
Net migration rate
Poverty rate
Log population
Proprietor share
Log density
Rurality
Birth rate
0
20
40
% increase in mean squared error
Importance indicates predictive reliance, not direction or causality.

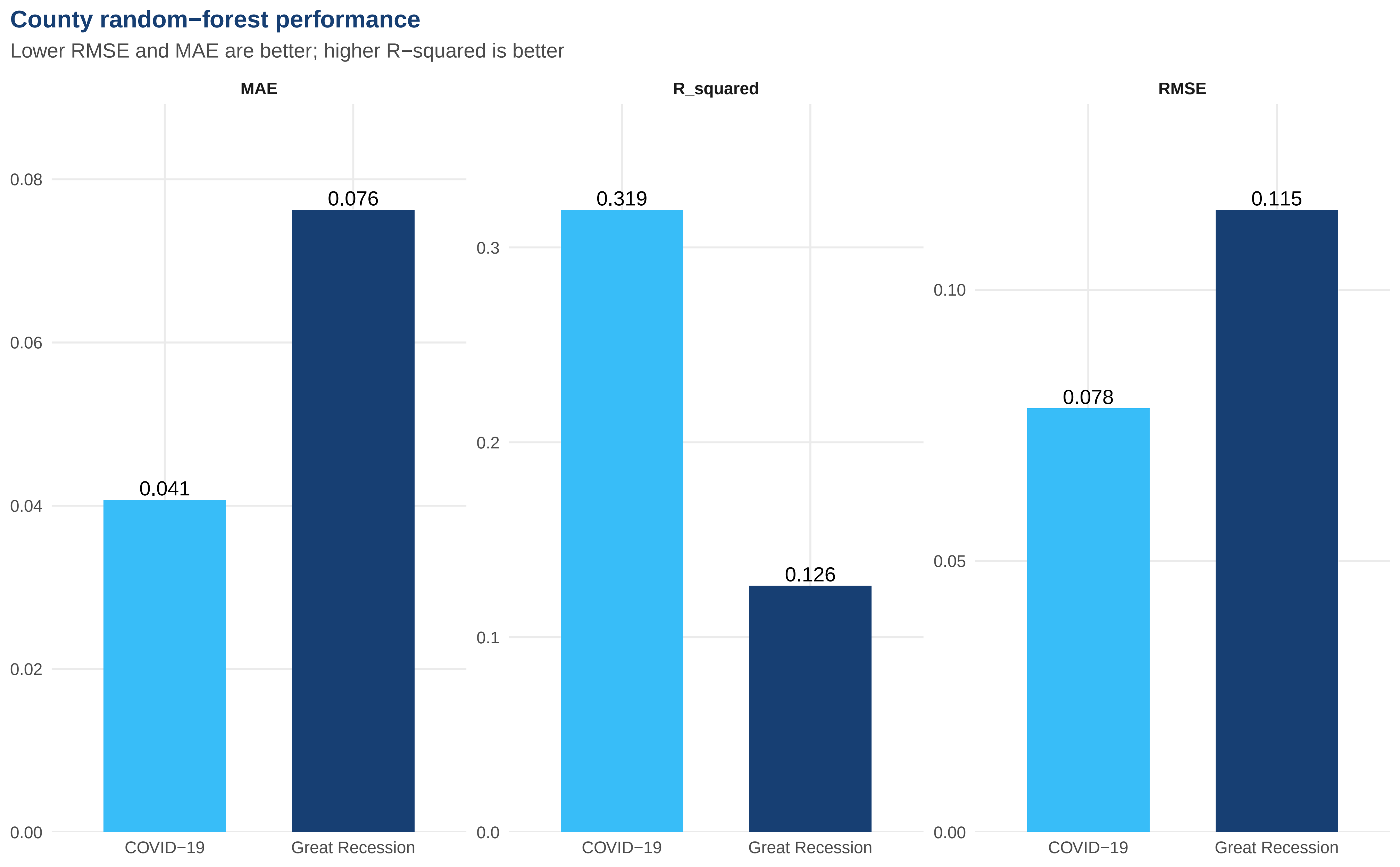
County random−forest response profiles

Marginal prediction curves for the most important numeric predictors



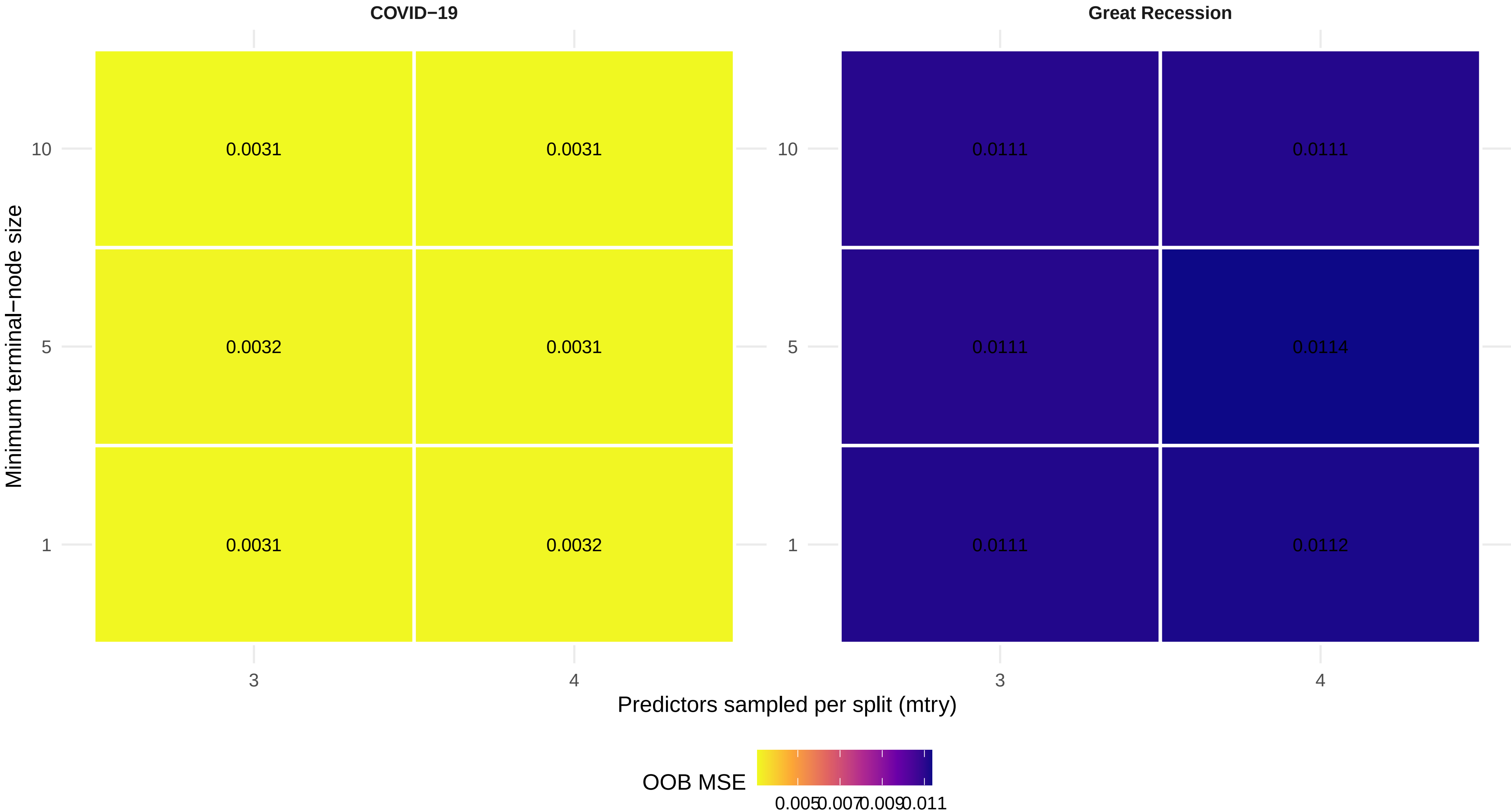
Curves describe model behavior and should not be interpreted causally.


County random−forest convergence
Out−of−bag error should stabilize as trees are added
COVID−19
Great Recession
Out−of−bag mean squared error
0.0050
0.0045
0.0040
0.0035
0.0200
0.0175
0.0150
0.0125
0
250
500
750
1000
Number of trees
COVID−19
Great Recession


County random−forest performance
Lower RMSE and MAE are better; higher R−squared is better
MAE
R_squared
RMSE
0.08
0.06
0.04
0.02
0.00
0.076
0.041
0.3
0.2
0.1
0.0
0.319
0.126
0.10
0.05
0.00
0.115
0.078
COVID−19
Great Recession
COVID−19
Great Recession
COVID−19
Great Recession

County random−forest hyperparameter search

Training−set out−of−bag MSE for combinations of mtry and node size



The final model uses the lowest−OOB−error combination.